



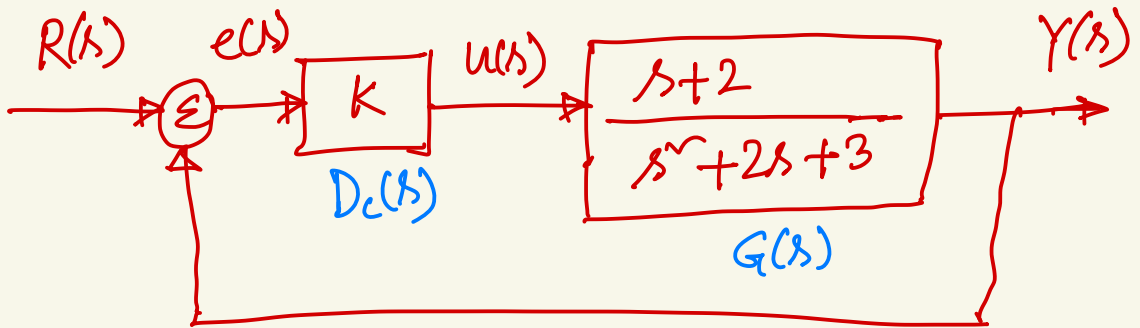
Root Locus method for control system design :-

Root locus is a graphical tool that shows how the poles of a closed-loop system moves as the control gain varies from zero to infinity.

This is especially helpful for analyzing higher order system.

This method can also help to analyze time-varying systems.

Let's consider the below system.



Here, the loop transfer function or the open loop transfer function is as follows

$$G_{\text{open}}(s) = \frac{K(s+2)}{s^2 + 2s + 3}$$

And the closed-loop transfer function is

$$T_u(s) = \frac{D_c(s) G(s)}{1 + D_c(s) G(s)}$$

$$T_u(s) = \frac{K(s+2)}{s^2 + (2+K)s + (3+2K)}$$

we can find the poles of the closed-loop transfer function by setting the denominator to zero.

$$s^2 + (2+K)s + (3+2K) = 0$$

we can now evaluate this equation for different values of K , and we can check how the poles or the roots of this equation move in the complex plane. The path traced out by the poles for

different values of k is called the root locus.

$$\text{ff } k=0, \quad s^2 + (2+0)s + (3+2 \cdot 0) = 0$$

$$\Rightarrow s^2 + 2s + 3 = 0$$

$\Rightarrow$ Open loop denominator

$$s_{1,2} = -1 \pm \sqrt{2}j = -1 \pm 1.4142j$$

$$\text{ff } k=1, \quad s^2 + 3s + 5 = 0$$

$$s_{1,2} = -1.5 \pm 1.65j$$

$$\text{ff } k=2, \quad s^2 + 4s + 7 = 0$$

$$s_{1,2} = -2 \pm 1.7j$$

$$\text{ff } k=3, \quad s^2 + 5s + 9 = 0, \quad s_{1,2} = -2.5 \pm 1.6j$$

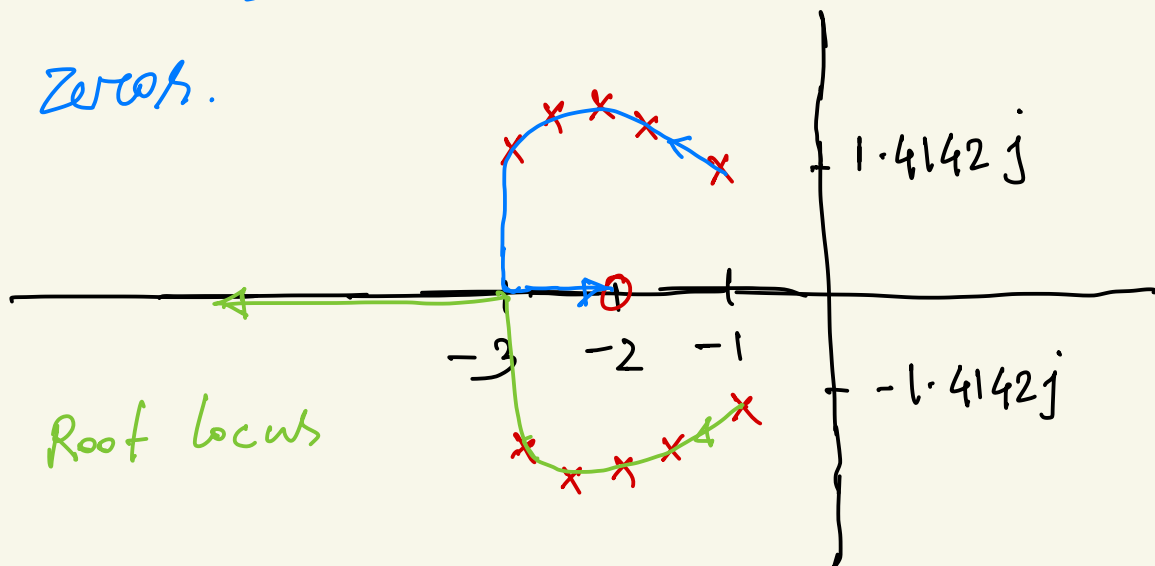
ff $k=4$, $s^2 + 6s + 11 = 0$, $s_{1,2} = -3 \pm 1.4j$

⋮

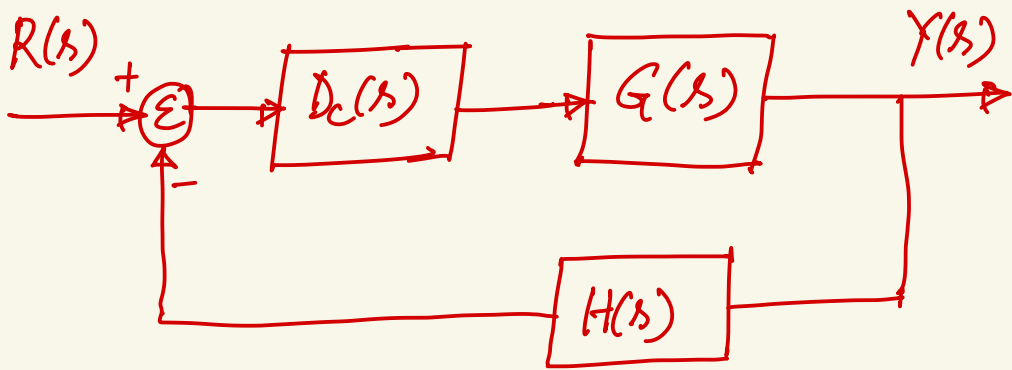
ff $k = \infty$, $ks + 2k = k(s + 2)$

$\Rightarrow$ zero of the open loop system.

Notice that we started at the poles and as we increase the value of k , we ended up in zeros.



This approach of finding the root locus is the brute force approach. Now let's formalize the above observation.



$$T_{cl}(s) = \frac{D_c(s) G(s)}{1 + D_c(s) G(s) H(s)}$$

The characteristic equation is

$$1 + D_c(s) G(s) H(s) = 0$$

$$G_{\text{open-loop}}(s) = D_c(s) G(s) H(s)$$

$$1 + G_{\text{open-loop}}(s) = 0$$

$$\Rightarrow 1 + K L(s) = 0$$

$$G_{\text{open-loop}}(s) = K L(s)$$

$1 + K L(s) = 0$ is called the root locus form.

$$L(s) = \frac{b(s)}{a(s)}$$

$$\Rightarrow 1 + K \cdot \frac{b(s)}{a(s)} = 0$$

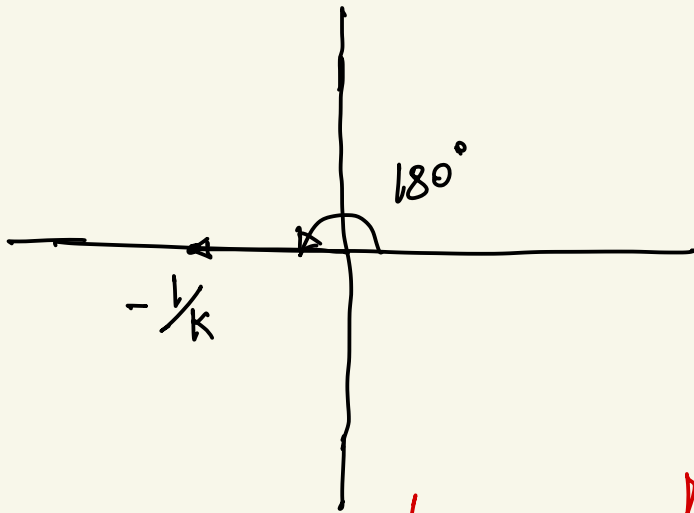
$$\Rightarrow a(s) + K \cdot b(s) = 0$$

ff $K=0$, $a(s)=0$, $\rightarrow$ poles

$$\Rightarrow \frac{a(s)}{k} + b(s) = 0$$

if $k \rightarrow \infty$, $b(s) = 0$, zeros.

Additionally, $L(s) = -\frac{1}{k}$



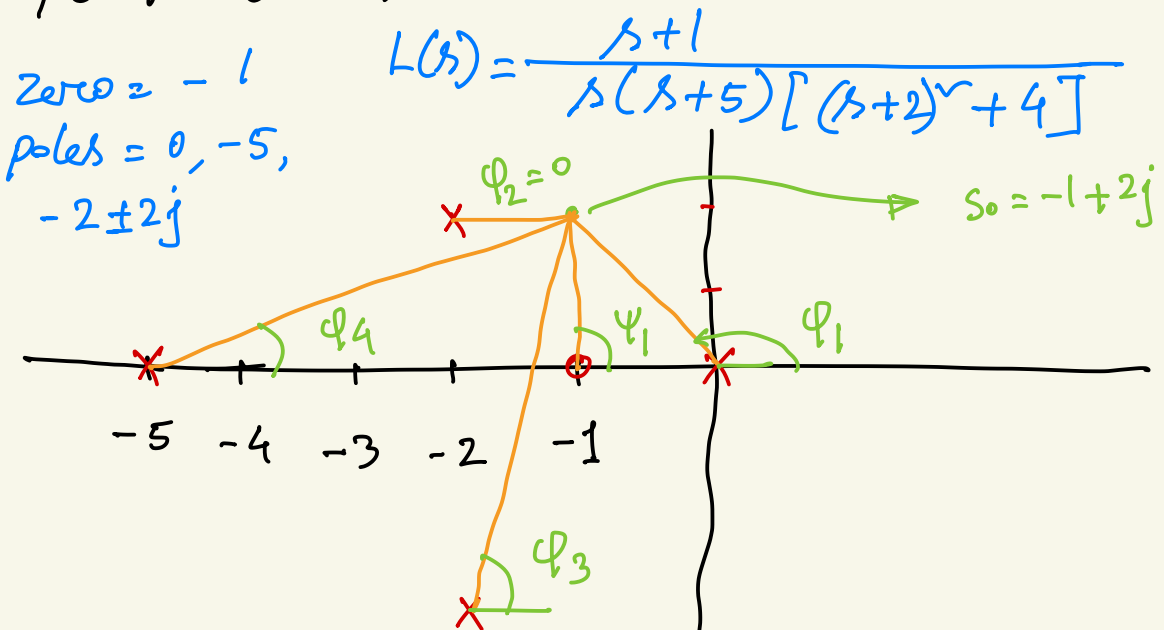
$L(s)$ is a complex number which has a magnitude and an angle

$$|L(s)| = \frac{|b(s)|}{|a(s)|} = \frac{1}{k}$$

$$\angle L(s) = \angle b(s) - \angle a(s) = 180^\circ + 360^\circ l$$

where, $l = 0, 1, 2, 3, \dots$

If a point satisfies the angle requirement, then it is on the root locus. Let's see an example:



$$\angle L(s) = \sum \psi_i - \sum \phi_i = 180^\circ$$

$$\Rightarrow \psi_1 - \phi_1 - \phi_2 - \phi_3 - \phi_4 = 90^\circ - 16.6^\circ - 0^\circ - 76^\circ - 26.6^\circ = -129.2^\circ$$

$\neq 180^\circ$

As a result, the s.o. point is not on the root locus. It is sometimes difficult to check for each point if it is on the root locus or not. So we will learn some rules to manually plot the root locus in an approximate but quicker way.

